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BSIT 4A

ITEP 414: System Administration and Maintenance
Week 3: Enterprise Server Deployment and Operating System Installation

UBUNTU SERVER 26.04 LTS INSTALLATION AND VERIFICATION GUIDE

Course / Subject: System Administration and Maintenance

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Hostname: server01

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1. PROJECT OVERVIEW

This document outlines the step-by-step installation, environment setup, and baseline verification of an Ubuntu Server 26.04 LTS virtual machine deployed via Oracle VM VirtualBox. It details headless server provisioning, guided LVM storage configuration, OpenSSH remote access setup, command-line system verification, and comparative research on server operating systems and firmware architectures.

2. OBJECTIVES

- Deploy a fully functional Linux server environment inside a hypervisor (Oracle VM VirtualBox).
- Configure non-root administrative credentials (sofialorraine) and set system domain hostname (server01).
- Implement guided LVM storage partitioning schemas for dynamic volume management.
- Enable secure remote administration through OpenSSH server service configuration.
- Validate core networking, package repositories, and service health using Linux CLI utilities.
- Compare enterprise server operating systems and low-level system firmware standards.

3. SOFTWARE REQUIREMENTS

- Hypervisor: Oracle VM VirtualBox (v7.0+)
- Operating System ISO: Ubuntu Server 26.04 LTS (Live Server 64-bit)
- Terminal Client: Built-in VirtualBox Console / SSH Terminal

4. VIRTUAL MACHINE SPECIFICATIONS

Parameter	Specification
VM Name	Ubuntu-Server-Week03
Guest OS Type	Linux / Ubuntu (64-bit)
Base Memory (RAM)	2048 MB (2 GB)
Processors	1 vCPU
Virtual Disk Type	VDI (VirtualBox Disk Image)
Storage Allocation	40.00 GB (Dynamically Allocated)
Network Adapter 1	NAT (Network Address Translation)

5. STEP-BY-STEP INSTALLATION PROCEDURE

Step 1: VM Creation & Media Attachment

1. Launch Oracle VM VirtualBox and select "New".
2. Name the virtual machine "Ubuntu-Server-Week03", select "Linux" as the type, and "Ubuntu (64-bit)" as the version.
3. Allocate 2048 MB RAM and 1 vCPU.
4. Create a virtual hard disk using VDI format, set to Dynamically Allocated with a capacity of 40.00 GB.
5. Attach the Ubuntu Server 26.04 LTS ISO file to the virtual optical drive and start the VM.

Step 2: Language & Keyboard Setup

1. Select "English" as the primary installer language.
2. Choose "English (US)" for both the keyboard layout and variant.

Step 3: Network & Proxy Configuration

1. Accept automatic IPv4 configuration assigned via DHCP on interface enp0s3.
2. Leave the Proxy Address field blank for a direct connection.
3. Accept the default regional Ubuntu Archive Mirror address (<http://ph.archive.ubuntu.com/ubuntu>).

Step 4: Storage Partitioning

1. Choose "Guided - Use An Entire Disk" targeting the 40 GB virtual drive (VBOX_HARDDISK).
2. Confirm the guided LVM layout (ubuntu-vg) and proceed past the destructive formatting prompt.

Step 5: Profile Configuration

1. Enter Full Name: Sofia Lorraine Gonzaga
2. Enter Server Name: server01
3. Set Username: sofialorraine
4. Create and confirm a secure administrative password.

Step 6: SSH & Optional Package Setup

1. Check the box for "[X] Install OpenSSH Server" and enable password authentication.
2. Skip optional snaps on the Featured Server Snaps screen.
3. Let the core installer run security updates, then select "Reboot Now" and unmount the ISO.

6. CONFIGURATION DETAILS

Parameter	Assigned Value / Setting
Hostname	server01
Administrative User	sofialorraine
Partition Scheme	Guided LVM (ubuntu-vg)
Root File System	ext4 (18.996 GB Volume)
Boot Partition	/boot (ext4, 2.00 GB)
Active Network Interface	enp0s3 (NAT Mode)
Assigned IPv4 Address	10.0.2.15/24
Enabled Services	OpenSSH Server (sshd)

7. VERIFICATION COMMANDS

After booting into the installed operating system, execute the following CLI commands to verify system state:

1. System Login Verification

Command: server01 login: sofialorraine

2. Verify Hostname

Command: hostname

Expected Output: server01

3. Verify Network IP Allocation

Command: ip addr

Expected Output: Interface enp0s3 inet 10.0.2.15/24

4. Test Outbound Internet Connectivity

Command: ping -c 4 google.com

Expected Output: 4 packets transmitted, 4 received, 0% packet loss

5. Update Package Repositories and Upgrade Packages

Commands:

```
sudo apt update  
sudo apt upgrade -y
```

6. Verify OpenSSH Daemon Status

Command: systemctl status ssh

Expected Output: Active: active (running)

8. WINDOWS SERVER INSTALLATION SUMMARY

Deploying Windows Server 2022/2026 involves distinct operational overheads compared to a headless Ubuntu Server installation:

- **Installation Footprint:** Requires significantly larger ISO installation media (~5 GB) and higher baseline disk space overhead (minimum 32 GB requirement for the OS volume).
- **Interface Options:** Supports both Server Core (lightweight CLI environment) and Server with Desktop Experience (Full GUI).
- **Identity Infrastructure:** Integrates natively with Active Directory Domain Services (AD DS) during post-installation wizard setup.
- **Hardware Demand:** Requires higher baseline RAM allocation (minimum 2 GB; recommended 4 GB to 8 GB) compared to headless Linux CLI setups (~512 MB to 1 GB).

9. OS COMPARISON TABLE

Criteria	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial (Per-Core)	Open Source (Free)	Open Source (Free)
User Interface	GUI or Server Core	Command Line (CLI)	Command Line (CLI)
Package Manager	winget / PowerShell	apt / snap	dnf / rpm
Default Security	Defender / AD	AppArmor / UFW	SELinux / FirewallD
Primary Use Case	Enterprise AD / IIS	Web, Cloud, Containers	Data Center / RHEL

10. PROBLEMS ENCOUNTERED

1. **Installer CPU Soft Lockup Warnings:** During background package downloading, console warnings appeared ("watchdog: BUG: soft lockup - CPU#0 stuck!").
2. **Package Mirror Retrieval Failures:** Running full system upgrades occasionally resulted in HTTP fetch errors for specific repository archives ("linux-firmware-amd-misc").

11. SOLUTIONS IMPLEMENTED

1. CPU Queuing Mitigation: Allowed the hypervisor background process sufficient execution time to complete security updates without forcibly interrupting or powering off the VM.
2. Fixing Missing Package Sources: Refreshed package source indices and executed the missing file flag using:

```
sudo apt update
```

```
sudo apt upgrade --fix-missing -y
```

12. LESSONS LEARNED

1. Headless OS Performance Efficiency: Running Linux Server without a desktop environment minimizes RAM/CPU usage, maximizing available host resources for server services.
2. Hypervisor NAT Architecture: Using default VirtualBox NAT mode allows full outbound Internet access without requiring manual proxy configurations.
3. Storage Flexibility with LVM: Logical Volume Management provides key advantages over static partitions, allowing administrators to expand disk volumes dynamically as storage demands grow.

13. REFERENCES

- Canonical Ltd. (2026). Ubuntu Server Guide. Ubuntu Documentation. <https://help.ubuntu.com/>
- Oracle Corporation. (2026). Oracle VM VirtualBox User Manual. <https://www.virtualbox.org/manual/>
- Tanenbaum, A. S., & Bos, H. (2015). Modern Operating Systems (4th ed.). Pearson.